# Medical Image Segmentation with a Vision Transformer: Swin UNETR on CT Spleen, with Attention-Based Explanation

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**Abstract.** We segment the spleen from 3D abdominal CT using a Swin-Transformer-based network (MONAI SwinUNETR, 62.2M parameters) on the Medical Segmentation Decathlon [11] Task09 dataset (41 volumes, 32 train / 9 validation), and we use the network's own attention layer to explain its decision. Our best model reaches a validation Dice of **0.9465**, within roughly one point of published results (Swin UNETR 95.79, nnU-Net 96.34) obtained with far more data and compute. The project's core is a controlled experimental process: a plausible hypothesis — that a learning-rate schedule and data augmentation would both help — was tested and **partially falsified**. Augmentation consistently *hurt* accuracy on this small dataset; a schedule bought training stability. An epoch-matched control isolates augmentation as the single cause of a ~5.4-point Dice loss. For explainability we define an attention-on-spleen ratio (1.47–2.33 across all nine cases), validate it with a two-armed null control (learned, not an artefact) and an occlusion faithfulness test (the prediction depends on the attended region, p=0.004). We report the analysis honestly, including validation-selection leakage and a single split as the main limitations.

## Requirement coverage

| Requirement | Where it is addressed in this report |
|---|---|
| **(a) Subject, goal, challenges** | §1 |
| **(b) Related work + innovation** | §2 |
| **(c) Implementation "with process" (what failed, how analysed)** | §5–§6 |
| **(d) >1 attempt, learning curves, hyper-parameter scan, a failed run analysed & defended** | §6 (Fig. 3), §7 |
| **(e) Own ideas compared with others' results** | §8 |
| **Project #3: use the attention layer to explain the decision** | §8 |

## 1. Subject, Goal, and Challenges

*Semantic segmentation* labels every voxel of a scan as belonging to a target structure or to background. Our goal is twofold, matching the assignment: (i) segment the spleen in 3D abdominal CT with a vision transformer, and (ii) use the model's attention to explain *where it looked* to reach its decision. Segmentation underpins real clinical tasks — organ-volume measurement, consistent quantification, and treatment planning — and in medicine a prediction is only trustworthy if it is also interpretable, which is why the explanation half is essential rather than decorative.

The main challenges were: **3D data volume** (a full CT volume does not fit in GPU memory, so we train on $96^3$ patches); **severe class imbalance** (the spleen is a tiny fraction of the scan); **very little labelled data** (41 volumes total), which shapes every later decision; and **tight compute** (a single NVIDIA L4 GPU with sessions capped at ~100 minutes). We deliberately chose the spleen — a small, fast organ — so that a single training run finishes in about an hour, letting us run several experiments; the course rewards process, not leaderboard rank.

## 2. Related Work and Our Innovation

Segmentation architectures descend from **U-Net** [1] (2015), an encoder–decoder with skip connections, and its 3D successors 3D U-Net and V-Net [2, 3] (2016). The strongest self-configuring baseline today is **nnU-Net** [4] (2021), which reports 96.34 Dice on the spleen. In parallel, transformers reached vision: **ViT** [5] (2021) brought self-attention to images at quadratic cost, and **Swin** [6] solved that with shifted local windows and linear complexity. From these came TransUNet, UNETR [7], and finally **Swin UNETR** [8] (2022), which pairs a Swin encoder with a U-Net-style decoder and reports 95.79 on the spleen. For explanation, prior work includes Grad-CAM [9] (2017) for CNNs and attention roll-out [10] (2020) for transformers.

**Our innovation is two-fold.** First, a controlled ablation showing that standard augmentation *lowers* accuracy on a 32-volume CT set — a result the large-scale papers do not surface because they are not compute-constrained. Second, a quantitative **attention-on-spleen ratio**, hardened into falsifiable evidence with a two-armed null control and an occlusion faithfulness test, rather than a single hand-picked heatmap, engaging directly with the attention-as-explanation debate [13, 14].

## 3. Data and Preprocessing

We use Medical Segmentation Decathlon Task09 (Spleen): 41 labelled 3D CT volumes, split 32 train / 9 validation. Each volume passes an identical pipeline: **orientation** to a canonical RAS frame; **resampling** to a fixed 1.5×1.5×2.0 mm voxel spacing (so one voxel means the same physical size in every scan); intensity **windowing** to the [-57, 164] HU soft-tissue range (discarding bone and air so capacity is spent on the organ); foreground cropping; and extraction of $96^3$ patches (four per step), sampled to be centred on the spleen to counter class imbalance.

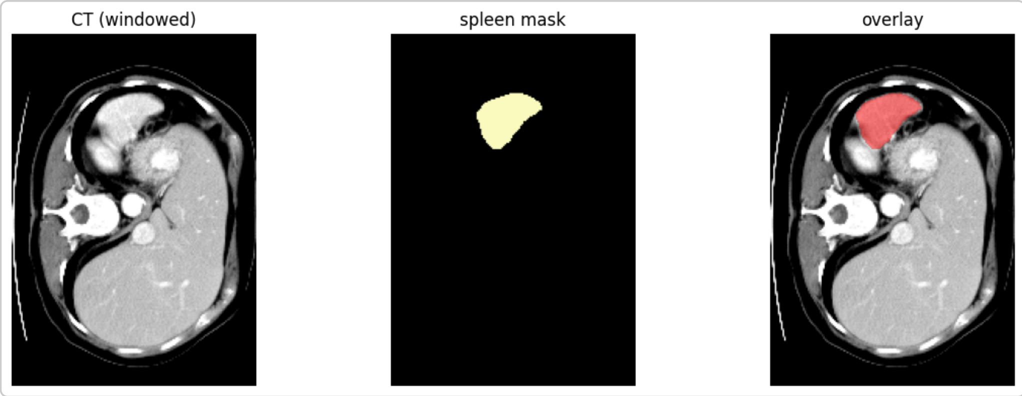


*Fig. 1. A CT slice from the 3D volume with the ground-truth spleen mask overlaid — the input and the reference the model is trained against.*

## 4. Model: Swin UNETR

We use MONAI's [12] **SwinUNETR**: a Swin-Transformer encoder (shifted-window self-attention over hierarchical, multi-scale stages, giving linear complexity and long-range context) feeding a convolutional U-Net decoder with skip connections that restore spatial detail. The configuration is 1 input channel (greyscale CT), 2 output channels (background / spleen), feature size 48, and 62.2M parameters.

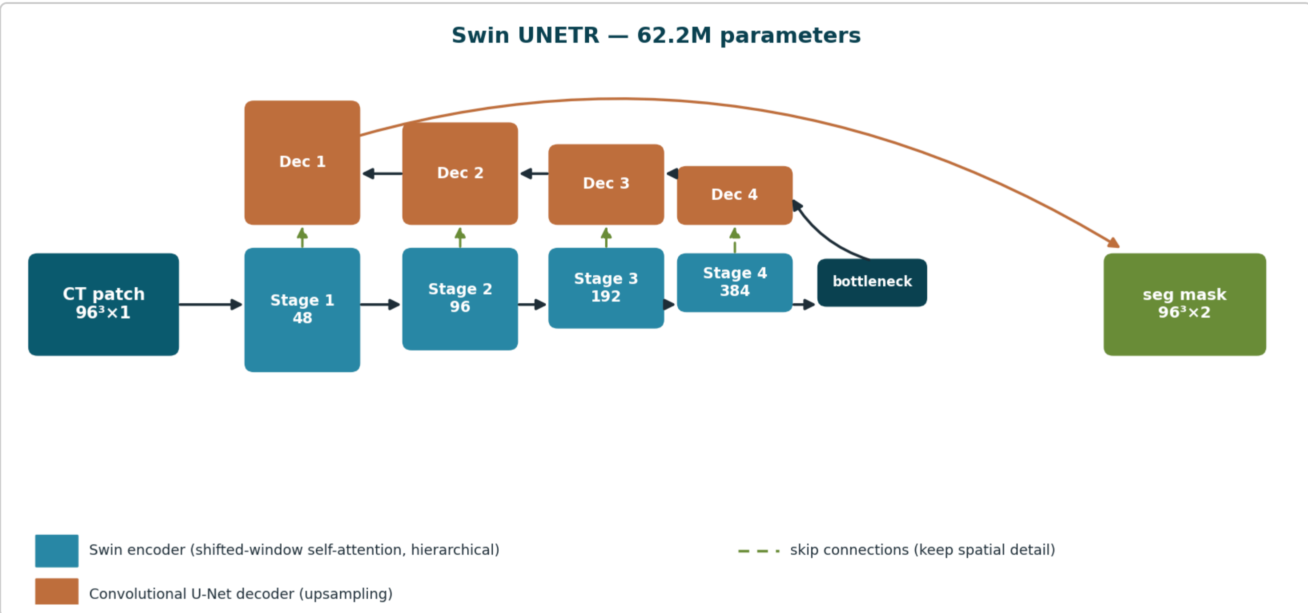


*Fig. 2. Swin UNETR: a CT patch is encoded by hierarchical shifted-window self-attention, passed through a bottleneck, and decoded by a convolutional U-Net decoder; skip connections carry spatial detail from encoder to decoder.*

## 5. Training Setup

Loss: DiceCELoss (Dice + cross-entropy). Optimizer: AdamW (lr 1e-4, weight decay 1e-5) with mixed-precision training to fit memory. Validation every five epochs via sliding-window inference. Hardware constraints shaped the runs directly: a batch size of two exhausts GPU memory, so we use one (four patches per step), and the ~100-minute session cap bounds each run to 60–100 epochs; every run checkpoints its best model to disk every five epochs. These are engineering realities we report as part of the process, not incidental details.

## 6. Experiments and Process

The heart of the project is a four-run scan in which we change one thing at a time — and one honest mistake where we did not.

- **Baseline** (fixed LR, no augmentation, 60 ep): Dice 0.9437. The validation curve was unstable — dips to 0.6359 (ep 10) and 0.7128 (ep 30) — and peaked on the final epoch, suggesting undertraining.
- **Hypothesis:** a cosine LR schedule with warm-up would stabilise training, and augmentation would add data variety and raise accuracy.
- **Exp1** (cosine + warm-up + augmentation, 80 ep): Dice 0.9088 — it *dropped*. This surprised us, and it exposed a methodological error: three variables changed at once, so no single one can be blamed.
- **Exp2** (adds an LR floor of 1e-5, 100 ep): Dice 0.9108. The floor was meant to fix undertraining from the decaying LR; it changed almost nothing, **falsifying the undertraining explanation** and leaving augmentation as the suspect.
- **Exp3** (same schedule + floor, augmentation removed, 70 ep): Dice 0.9465 — the chosen run. It differs from exp2 by exactly one variable, so exp3-vs-exp2 isolates augmentation, and exp3-vs-baseline (both no-aug) isolates the schedule. Conclusion: the schedule bought *stability*; removing augmentation bought *accuracy*.

**Epoch-matched control.** Because exp2 ran 100 epochs and exp3 only 70, we re-ran the augmented recipe at exactly 70 epochs: Dice 0.8927 (it peaked at epoch 40 and then plateaued, so it converged rather than undertrained). At a matched budget the single-variable cost of augmentation is **~5.4 Dice points** — larger than the unmatched 3.5 — so controlling the confound strengthened the finding. The likely culprit is RandRotate90 (90° rotations of an abdomen are anatomically impossible, forcing invariance to inputs never seen at test), but as the augmentations were applied as a bundle this remains a hypothesis, not a measured attribution.

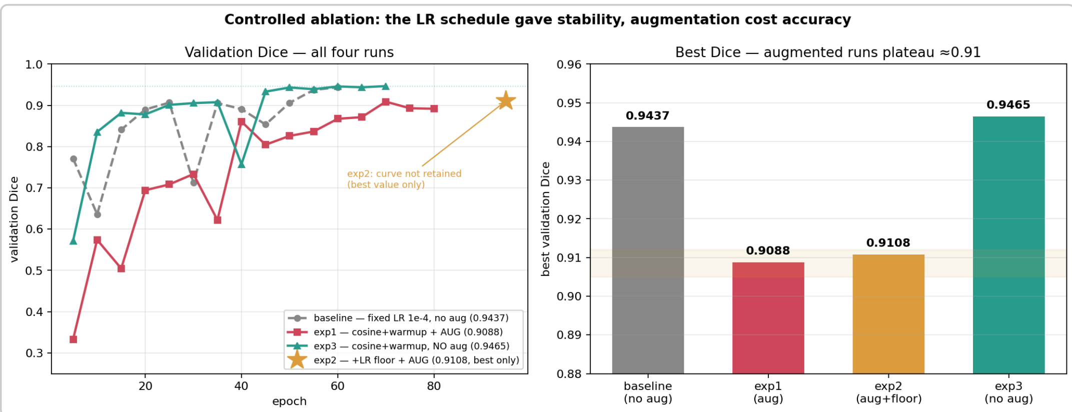


*Fig. 3. Validation Dice across the four runs. exp3 (no augmentation) is highest and smoothest; exp2 is shown as a single starred point (only its best score was retained).*

## 7. Statistical Rigor

Because the validation set is only nine volumes, we quantified the noise rather than asserting it. **Seeds:** exp3 retrained with three random seeds gives 0.9465 / 0.9508 / 0.9457, mean 0.9477 ± 0.0028 — a spread as large as the 0.003 gap over the baseline, so that gap is within noise. **Significance:** a paired Wilcoxon signed-rank test (exp3 vs baseline, per-volume) gives mean difference +0.0028, 95% CI

[−0.0004, +0.0065], p=0.20 — not significant; we therefore claim exp3 is the *more stable* choice, not the more accurate one. **Per-volume:** exp3 ranges 0.9149–0.9655 across the nine volumes (worst = volume 0, an over-segmentation on a low-contrast boundary) and beats the baseline on five of nine — confirming the set is noisy. **Boundary metrics:** HD95 3.95 mm, Surface-Dice 0.889 (2 mm), precision 0.944, recall 0.950.

## 8. Explainability and Our Contribution

To explain the decision we hook the softmax of the deepest WindowAttention layer, average over its 24 heads and over queries, reshape the result to the $6^3$ window grid, upsample to $96^3$, and overlay it on the CT. We turn this into a falsifiable number — the **attention-on-spleen ratio** (mean attention inside the spleen mask divided by mean attention outside it) — and measure it on all nine validation cases: it ranges 1.47–2.33 and exceeds 1 in every case.

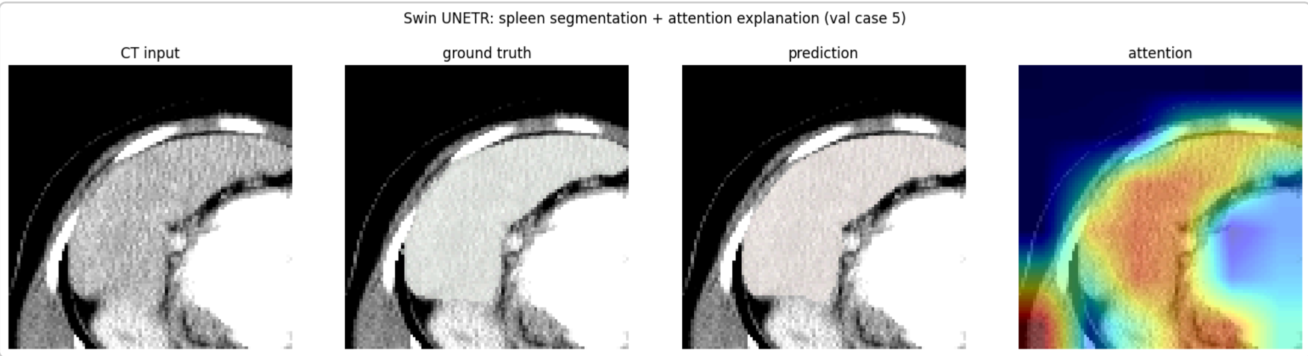


*Fig. 4. One validation case: CT input, ground-truth mask, model prediction, and the attention heatmap the model used.*

A ratio above 1 could, however, be an artefact of patches being centred on the spleen. We rule this out with two controls, and then test faithfulness (Fig. 5). **Null control:** the trained model gives ratio 1.90 on the true spleen, versus 1.11 for an untrained (random-weight) model on the same mask and 1.28 for the trained model on random off-spleen boxes — so the concentration is *learned*, not initialisation or centred-crop bias. **Occlusion:** zeroing the top-15% highest-attention voxels collapses Dice by 0.44, versus 0.015 for a random equal-size set and 0.000 for the lowest-attention set (paired Wilcoxon p=0.004) — the prediction genuinely *relies* on the attended region. **Layer choice:** measuring all eight layers shows the on-spleen ratio rises with depth (early layers ≤1.24, the last three 1.7–2.2), justifying the deep hook. We stop short of claiming causation beyond localisation, consistent with the open debate [13, 14].

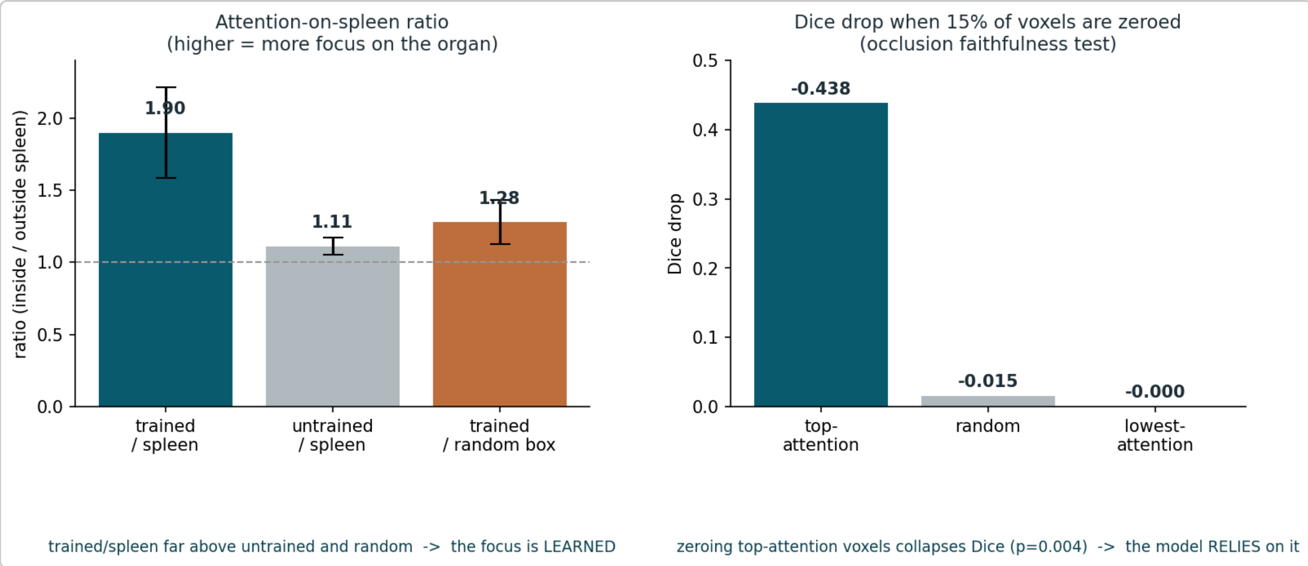


*Fig. 5. The two strongest explanation results. Left: the attention-on-spleen ratio is far higher for the trained model than for an untrained model or random regions, so the focus is learned. Right: zeroing the highest-attention voxels collapses Dice while random/lowest-attention occlusion does not, so the decision depends on the attended region.*

## 9. Comparison to Published Work

Our best Dice, 94.65, sits about one to two points below the Swin UNETR paper (95.79) and nnU-Net (96.34). The gap tracks their far larger training data, five-fold cross-validation, model ensembling, heavy per-task tuning, and much greater compute — not a broken pipeline. On the boundary metrics that medical papers report, our numbers (HD95 3.95 mm, Surface-Dice 0.889) are competitive. The assignment does not require beating published systems; our contribution is the controlled augmentation-cost finding and the falsifiable attention-explanation methodology, both validated above.

## 10. Limitations and Future Work

- **Validation-selection leakage.** The schedule, warm-up, LR floor, and augmentation setting were all chosen on the same nine validation volumes we then report, with no held-out test set, so every number is optimistic by an unknown amount. This is the main weakness.
- **Single split.** All results rest on one 32/9 split; the three seeds vary only initialisation, not the split. A held-out test set and k-fold cross-validation would remove both caps (a k-fold script is included with the submission).
- **Bundle augmentation.** We show the bundle hurts, not which transform; a per-transform (leave-one-out) ablation is future work.
- **Best-at-final-epoch.** Some runs peaked on their last epoch (the session cap, not convergence), so comparisons are at a fixed budget rather than at saturation.

## 11. Conclusion

We built a single-GPU Swin UNETR that segments the spleen at Dice 0.9465, within a point of published systems, and we explain its decisions with a quantitative, controlled attention analysis. The project's real value is the process: we formed a reasonable hypothesis, tested it, found it wrong, and isolated the true cause one variable at a time — concluding, against our own expectation and confirmed by an epoch-matched control, that standard augmentation hurts on this small dataset. Throughout, we report the limitations plainly. The central lesson we take is to change one variable at a time and to make every claim falsifiable.

## 12. References


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